

Game Flow

1. The player starts the game and enters the opening screen.
2. The game displays a welcome message with the title: **CiteGuard VR: Space Mission AI Assistant**.
3. The welcome screen briefly explains that the player will learn how an AI system answers questions using trusted evidence instead of guessing.
4. The player is given two choices: **Start Tutorial** or **Skip Tutorial**.
5. If the player chooses the tutorial, a set of tutorial cards appears in front of them.
6. The first tutorial card introduces **RAG**, explaining that Retrieval-Augmented Generation means the AI retrieves useful information before generating an answer.
7. The second tutorial card explains **User Query**, showing that every mission starts with a question from a user or crew member.
8. The third tutorial card explains **Evidence Cards**, showing that the AI needs trusted information to answer correctly.
9. The fourth tutorial card explains **Retrieval**, where the player finds evidence that matches the query.
10. The fifth tutorial card explains **Reranking**, where the player orders evidence from strongest to weakest.
11. The sixth tutorial card explains the **LLM Panel**, where the AI uses the selected evidence to generate an answer.
12. The seventh tutorial card explains **Validation**, where the answer is checked for grounding, citation support, and hallucination risk.
13. The final tutorial card tells the player that wrong or weak evidence can cause the AI to generate an incorrect or unsupported answer.
14. After the tutorial, or if the player skips it, the player reaches the game selection screen.
15. In the first version, the only available game is **Space Mission AI Assistant**.
16. The player selects **Space Mission AI Assistant** and enters a spaceship command center.
17. The command center includes a main mission screen, an evidence box, a reranking station, an LLM panel, a validation station, and a score display.
18. A role briefing appears on the main screen.
19. The player is told: **“You are the AI Mission Officer.”**
20. The game explains the player’s duty: help the spaceship crew answer mission-critical questions using only reliable evidence.
21. The player is reminded that their goal is not just to get an answer, but to produce a trustworthy answer.
22. The first mission begins when a crew query appears on the command screen.
23. Example crew query: **“Can we activate the solar shield during low battery mode?”**
24. The player reads the query and thinks about what information is needed to answer it.
25. The Evidence Box becomes active.
26. The player opens the Evidence Box.
27. The Evidence Box displays 4 to 5 short evidence cards connected to that specific query.
28. Each evidence card contains only a few sentences, so the player can quickly understand it.
29. For the solar shield query, the evidence cards may include information about battery level, solar shield requirements, low battery mode, backup systems, and unrelated ship functions.

30. The player examines the cards and decides which ones are useful for answering the crew query.
31. Some evidence cards are directly useful, some are only partially related, and some are distractors.
32. The player selects the evidence cards they believe are most relevant.
33. The player drags the selected evidence cards to the Reranking Station.
34. At the Reranking Station, the player places the evidence cards in order from best match to weakest match.
35. The strongest evidence should be placed first because the AI answer depends heavily on the best-ranked information.
36. For the example query, the best evidence would be the card saying the solar shield requires battery above 40% and the card saying low battery mode begins below 25%.
37. After ranking the evidence, the player sends the ranked evidence to the LLM Panel.
38. The LLM Panel generates an answer based on the evidence provided by the player.
39. If the player selected and ranked the correct evidence, the LLM Panel produces a strong grounded answer.
40. Example strong answer: **“No. The solar shield should not be activated during low battery mode because the shield requires battery above 40%, while low battery mode begins below 25%.”**
41. If the player selected some useful evidence but missed a key card, the LLM Panel produces an incomplete answer.
42. If the player selected weak or irrelevant evidence, the LLM Panel produces an answer with higher hallucination risk.
43. After the answer is generated, the player moves the answer to the Validation Station.
44. The Validation Station checks whether the answer is supported by the selected evidence.
45. The validation result shows the player how trustworthy the answer is.
46. The validation screen displays scores for retrieval quality, evidence ranking, grounding, citation support, hallucination risk, and overall trust.
47. If the answer is well supported, the game approves the answer and sends it to the crew.
48. If the answer is weak, incomplete, or unsupported, the game warns the player.
49. Example warning: **“High hallucination risk: The answer does not include the battery requirement needed to decide whether the solar shield can be activated.”**
50. The player receives feedback explaining what went right and what went wrong.
51. The player may be allowed to retry the same mission by selecting and ranking better evidence.
52. After completing the mission, the player moves to the next crew query.
53. Each new query repeats the same workflow: read query, open evidence box, choose evidence, rerank evidence, generate answer, validate answer, receive score.
54. As the game continues, the missions become slightly more difficult.
55. Later missions may include more similar-looking evidence cards, incomplete evidence, or tempting distractor cards.
56. The final mission asks the player to complete the full RAG workflow with minimal guidance.
57. After the final mission, the player receives a complete performance summary.
58. The final summary shows how well the player performed across all missions.

59. The summary includes total score, retrieval accuracy, ranking accuracy, grounding score, citation support score, hallucination prevention score, and final trust rating.
60. The game ends with a learning message: **“Trustworthy AI depends on trustworthy evidence. A RAG system is only reliable when the retrieved information supports the generated answer.”**
61. The player can then choose to replay the game, return to the tutorial, or exit.

This version explains the gameplay clearly without getting into technical implementation. It should make sense to your professor as a complete learning experience from start to finish.